



Request for Information Response

Artificial Intelligence (AI) Action Plan

Developed for:

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Request for Information (RFI) Response

CostQuest Associates (CQA) is pleased to provide this RFI response to the White House Office of Science and Technology Policy (OSTP) Request for Information on the Development of an Artificial Intelligence Action Plan.

CQA is a leading broadband/data network infrastructure consulting firm specializing in cost analysis, network modeling, mapping, regulation, and data science. CQA has actively integrated Agentic AI into our operations. This integration has led to innovations such as autonomous network optimization, where AI agents analyze real-time data to enhance network performance and efficiency, and predictive maintenance protocols, which utilize AI to forecast potential infrastructure failures, thereby minimizing downtime and extending asset lifespans.

We appreciate the opportunity to provide input and comments on the development of the Artificial Intelligence (AI) Action Plan as directed by the Presidential Executive Order on January 23, 2025. Our expertise in data-driven solutions and geospatial analytics positions us to offer insights into sustaining and enhancing America's AI leadership while fostering private-sector innovation.

Below are a number of data-driven and evidence-based recommendations from our team. We would be pleased to discuss these recommendations further with OSTP team members, stand at the ready to aid in their implementation, and members of our team would be happy to be involved in future working groups and/or sessions of the group related to this plan.

Recommendations

1. Utilize AI for Predictive Maintenance of Data Network Infrastructure

AI can play a pivotal role in the predictive maintenance of data networks. By analyzing data from network sensors and performance metrics, AI models can forecast potential failures or degradations in service. This proactive approach allows for timely maintenance, reducing downtime, and improving service reliability for consumers.

2. Enhance Data Network Infrastructure Mapping Accuracy with AI



Leverage the power of AI-enhanced data network infrastructure maps to accelerate decision-making and optimize network planning. Maps, developed using advanced AI algorithms, provide highly accurate insights by analyzing geospatial data, satellite imagery, and infrastructure information. These AI-enhanced maps effectively identify serviceable areas, eliminating the need for costly and time-consuming manual assessments. Our work in developing the Broadband Serviceable Location Fabric (BSLF) has proven the value of AI-powered mapping in pinpointing serviceable locations with precision, enabling more efficient broadband and internet infrastructure expansion efforts. These tools can be utilized throughout the government to bring a single source of truth and eliminate waste, fraud, and abuse.

3. Leverage AI for Demand Forecasting and Network Optimization

Understanding and anticipating user demand is vital for network optimization. AI models can analyze historical usage data and predict future demand trends, enabling providers to adjust their networks accordingly. This foresight helps in maintaining quality of service and in making informed decisions about capacity upgrades. See #9 for more information/context.

4. Implement AI-Driven Cost Modeling for Network Expansion

Deploying data network infrastructure, especially in rural and underserved areas, requires careful financial planning. AI-driven cost modeling can analyze various factors such as terrain, population density, and existing infrastructure to provide accurate cost estimates for network expansion projects. This precision aids in the efficient allocation of resources and maximizes the impact of investment. Our work in cost modeling has underscored the benefits of integrating AI to enhance the accuracy of financial projections.

5. Advanced Cost Modeling for Optimal Data Center Placement

Building upon our proven cost modeling methodologies, CostQuest can assist states in identifying optimal locations for new data centers. By analyzing factors such as infrastructure costs, energy availability, and



proximity to end-users, our models can pinpoint sites that offer the best balance between operational efficiency and cost-effectiveness.

6. Refining Precision Agriculture Analytics through AI

Artificial intelligence (AI) is revolutionizing precision agriculture by analyzing vast datasets from IoT sensors, drones, and automated farming equipment to enhance efficiency and sustainability. AI-powered analytics enable real-time monitoring of soil conditions, weather patterns, pest activity, and crop health, allowing farmers to make data-driven decisions on when to irrigate, fertilize, or apply pest control measures—ultimately reducing waste and maximizing yields. By integrating IoT-enabled sensors, AI can track moisture levels, nutrient deficiencies, and disease outbreaks, sending automated alerts to farmers for targeted intervention.

Additionally, AI-powered livestock tracking systems monitor animal health, movement, and feeding patterns, ensuring optimal care, and reducing losses. Connected farm equipment with AI-driven GPS tracking improves fleet management, fuel efficiency, and predictive maintenance, minimizing downtime and operational costs. AI-driven precision agriculture not only enhances productivity but also supports sustainable farming practices by conserving water, optimizing fertilizer use, and reducing environmental impact. As AI continues to evolve, its integration with IoT technology and broadband-enabled farms will pave the way for smarter, more resilient agricultural ecosystems.

7. The Fabric: A Locational Anchor for AI-Driven Networks

The Fabric serves as a foundational locational anchor for AI-powered mesh networks, enabling seamless connectivity across distributed yet non-fixed infrastructures. As artificial intelligence continues to shape real-time data processing and automation, networked devices—including IoT sensors, LoRa-enabled smart systems, and AI-driven edge computing nodes—can be mapped to a Fabric anchor point, providing a structured, intelligent framework for connectivity.



Beyond broadband deployment, the Fabric unlocks AI-driven optimization by tying Broadband Serviceable Locations (BSLs) to next-generation networks. AI can analyze which BSLs are actively supporting advanced technologies, allowing for predictive network management, smarter resource allocation, and automated decision-making. This AI-powered analysis enhances network efficiency, infrastructure investment, and digital ecosystem resilience, ensuring that connectivity expands strategically to support the evolving demands of AI-driven applications.

8. Advancing Data Center Placement and Deployment in Rural Areas

Our analytical framework can be adapted to support construction firms, hyperscalers, commercial real estate investment firms, and others in assessing potential co-located data center sites by evaluating land use patterns, environmental impact, and logistical considerations, ensuring that new facilities are both sustainable and strategically positioned.

9. Enhancing the Opportunity Finder Tool for Data Center Site Selection

CostQuest's Opportunity Finder tool, designed to identify promising areas for data network infrastructure expansion, can be refined to evaluate data center placement opportunities. By incorporating criteria specific to data centers, such as energy grid capacity and fiber optic network proximity, the tool can provide actionable insights for strategic deployment.

10. Expanding Mobile Edge Compute Expertise to Larger Data Centers

Our experience with mobile edge compute centers, essentially miniature data centers, provides a solid foundation for advising on larger-scale data center projects. We can offer guidance on scaling these deployments, addressing challenges related to latency, data throughput, and integration with existing data network infrastructure.

11. Power Consumption Insights Based On CostQuest Data

As broadband serviceable locations (BSLs) increasingly rely on artificial intelligence (AI) for data processing, understanding their power



consumption needs is critical. CostQuest's data-driven power analysis identifies where electricity demand is highest, providing power grid experts with the insights needed to strategically allocate energy resources. AI-driven applications consume exponentially more power than traditional websites. For example, AI-powered models like ChatGPT require approximately 2.9 kilowatt-hours per 1,000 prompts, compared to 0.001 kilowatt-hours for a simple Google search. This emerging metric—kilowatts per token processed—is becoming a crucial measure in assessing the energy footprint of AI operations.

BSLs are not only consumers of AI but also generators of data that fuel AI advancements. By measuring the power consumption per unit of data processed, CostQuest's insights can guide the development of efficient, resilient, and sustainable power infrastructure. This approach protects national security, economic stability, and the integrity of the country's energy grid by ensuring responsible power distribution. Moreover, fortifying the grid against cyber threats is paramount. CostQuest's intelligence can help mitigate risks from malicious actors who may target data centers and critical infrastructure through mass outages or coordinated cyberattacks. By aligning broadband expansion efforts with power grid security, we contribute to a safer, more efficient digital and energy ecosystem for the future.

12. Develop AI-Powered Tools for Spectrum Management

Efficient spectrum management is essential for optimizing data network infrastructure services. AI can analyze spectrum usage patterns to identify underutilized frequencies and suggest optimal allocation strategies. This dynamic management ensures better utilization of available spectrum and can alleviate congestion in high-demand areas.

13. Ensure Data Privacy and Security in AI Applications

As AI systems process vast amounts of data, ensuring the privacy and security of user information is paramount. We recommend implementing robust encryption protocols and access controls. Additionally, AI models should be designed to anonymize data wherever possible to protect user identities.

14. Promote Collaboration Between AI and Broadband/Data Network Infrastructure Stakeholders

Fostering collaboration between AI researchers, broadband/data network infrastructure service providers, and policymakers can accelerate innovation. Establishing consortiums or working groups focused on AI applications in this sector can facilitate knowledge sharing and the development of best practices.

Data Privacy and Intellectual Property Considerations

1. Robust Data Privacy Protections

As AI systems process increasing amounts of sensitive information, it is imperative to establish stringent data privacy regulations. Protecting consumer data builds trust and ensures compliance with ethical standards.

2. Safeguarding Small Businesses Against IP Claims

We recommend implementing policies that protect small businesses from disproportionate intellectual property claims or theft. Providing legal support and establishing clear guidelines can prevent larger entities from leveraging IP laws to stifle competition.

3. AI Indemnification Frameworks

Developing indemnification policies specific to AI applications can shield businesses from liabilities arising from AI-driven decisions. Clear frameworks ensure that responsibility is appropriately assigned, fostering innovation while mitigating risk.

Conclusion

CostQuest Associates is committed to contributing to the advancement of AI technologies in the United States. We believe that the above recommendations align with the goals of the AI Action Plan and will help ensure America's continued leadership in AI. We welcome the opportunity to collaborate with OSTP to further refine these strategies and contribute our expertise to the development and implementation of effective AI policies.